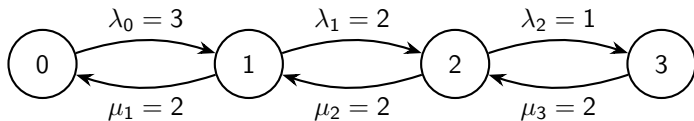


Practice Solution: CTMC Diagram and Balance

Given

$$\lambda_0 = 3, \quad \lambda_1 = 2, \quad \lambda_2 = 1, \quad \lambda_n = 0 \ (n \geq 3),$$
$$\mu_n = 2, \quad n = 1, 2, \dots$$

Only states 0, 1, 2, 3 can have positive steady-state probability.



Local balance equations:

$$3\pi_0 = 2\pi_1, \quad 2\pi_1 = 2\pi_2, \quad \pi_2 = 2\pi_3.$$

Practice Solution: Steady-State Probabilities

From local balance,

$$\pi_1 = \frac{3}{2}\pi_0, \quad \pi_2 = \pi_1 = \frac{3}{2}\pi_0, \quad \pi_3 = \frac{1}{2}\pi_2 = \frac{3}{4}\pi_0.$$

Use normalization:

$$\pi_0 + \pi_1 + \pi_2 + \pi_3 = 1.$$

Substitute the expressions in terms of π_0 :

$$\pi_0 + \frac{3}{2}\pi_0 + \frac{3}{2}\pi_0 + \frac{3}{4}\pi_0 = 1.$$

$$\frac{19}{4}\pi_0 = 1 \quad \Rightarrow \quad \pi_0 = \frac{4}{19}.$$

Therefore,

$$\pi_0 = \frac{4}{19}, \quad \pi_1 = \frac{6}{19}, \quad \pi_2 = \frac{6}{19}, \quad \pi_3 = \frac{3}{19}$$

and

$$\pi_n = 0, \quad n = 4, 5, \dots$$

Practice Solution: Effective Arrival Rate and Queue Size

The effective arrival rate is

$$\lambda_e = \sum_{n=0}^{\infty} \lambda_n \pi_n = 3\pi_0 + 2\pi_1 + \pi_2 = \frac{12}{19} + \frac{12}{19} + \frac{6}{19} = \frac{30}{19}.$$

Expected number in the system:

$$E[N] = \sum_{n=0}^3 n\pi_n = \frac{6}{19} + \frac{12}{19} + \frac{9}{19} = \frac{27}{19}.$$

For a single-server system, queue length in state n is $\max\{n-1, 0\}$. Thus,

$$E[N_Q] = (0)\pi_1 + (1)\pi_2 + (2)\pi_3 = \frac{6}{19} + \frac{6}{19} = \frac{12}{19}.$$

Practice Solution: Time Metrics

By Little's Law with λ_e ,

$$E[T] = \frac{E[N]}{\lambda_e} = \frac{27/19}{30/19} = 0.9$$

$$E[T_Q] = \frac{E[N_Q]}{\lambda_e} = \frac{12/19}{30/19} = 0.4$$

Final answers:

$$\pi_0 = \frac{4}{19}, \quad \pi_1 = \frac{6}{19}, \quad \pi_2 = \frac{6}{19}, \quad \pi_3 = \frac{3}{19}, \quad \pi_n = 0 \quad (n \geq 4)$$

$$\lambda_e = \frac{30}{19}, \quad E[N] = \frac{27}{19}, \quad E[N_Q] = \frac{12}{19}, \quad E[T] = 0.9, \quad E[T_Q] = 0.4$$